

GANESH MARMAT

Agentic AI Engineer | Distributed Systems | AI Automation

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PROFESSIONAL SUMMARY

Senior Software Engineer with 10+ years of experience designing and delivering scalable distributed systems, microservices architectures, and enterprise platforms. Over the last 3+ years, specialised in Agentic AI engineering, Large Language Model (LLM) integration, LangGraph-based AI agent development, and ML pipeline design. Proven ability to deliver AI-driven automation products end-to-end — from requirements gathering and system design through deployment and monitoring. Strong expertise in Python, .NET Core, REST API design, Elasticsearch, and cloud platforms including Azure and AWS.

TECHNICAL SKILLS

AI & Agentic Systems: Agentic AI, LangGraph, LLM Integration, Large Language Models, Prompt Engineering, AI Workflow Automation, Conversational AI, Tool Calling, OpenAPI, Swagger, FastAPI

Machine Learning: ML.NET, Binary Classification, Multiclass Classification, Data Pipelines, Model Training, Model Evaluation, Feature Engineering, Supervised Learning

Programming Languages: Python, C#, .NET Core (.NET 5 / 6 / 8), JavaScript, TypeScript

Web & API: REST API, FastAPI, ASP.NET Core, Angular, WebAPI, WCF, OData

Architecture & Design: Microservices, Distributed Systems, Event-Driven Architecture, System Design, Domain-Driven Design, MVVM, SOLID Principles

Data & Search: Elasticsearch, SQL Server, T-SQL, MongoDB, SQLite, Data Indexing, Query Optimisation

Cloud & DevOps: Microsoft Azure, AWS, CI/CD, Azure DevOps, TeamCity, Docker, WIX Installer

Tools & Platforms: Git, Visual Studio, JIRA, Bitbucket, TFS, NUnit, Selenium

WORK EXPERIENCE

Cloud Software Group (CSG) — via Arctera (formerly Veritas Technologies) | Pune, India

Senior Software Engineer | 03/2020 – Present

Agentic AI & LLM Engineering

- Architected and delivered a production-grade Agentic AI chatbot using Python and LangGraph, enabling intelligent multi-step task execution and workflow automation across enterprise services.
- Engineered an Agentic QA Framework (LangGraph, Python, FastAPI) that empowers QA engineers to define API test cases in plain English; framework auto-discovers and dynamically invokes tools from OpenAPI/Swagger JSON — eliminating manual scripting.
- Designed and implemented AI workflows integrating LLM-based reasoning with backend REST APIs and enterprise services, enabling end-to-end business process automation.
- Applied prompt engineering techniques to optimise LLM outputs for accuracy, reliability, and task-specific performance.
- Designed modular, extensible AI agent architecture supporting rapid addition of new tools, integrations, and capabilities.

Machine Learning & AI Systems

- Developed an Intelligent Email Review System using ML.NET (4 years): binary classification model to filter relevant emails and multiclass classification model to categorise emails into business-defined categories.
- Delivered interactive bar-chart data visualisations surfacing ML classification results to business stakeholders for decision-making.
- Built end-to-end ML data pipelines covering data ingestion, preprocessing, feature engineering, model training, evaluation, and deployment.

- Automated previously manual email review workflows using ML-based decision systems, reducing processing time and operational overhead.

System Design & Backend Engineering

- Designed microservices-based distributed system architecture for enterprise-scale .NET Core applications.
- Developed high-throughput REST APIs processing large-scale data workloads with optimised performance.
- Built Elasticsearch (v7.10) search and audit system for compliance monitoring of communication data sources including email and Bloomberg.
- Optimised system performance through strategic indexing, query tuning, caching, and profiling.
- Implemented OData-based data extraction layer enabling flexible, standards-based querying from enterprise servers.

Platform Engineering & DevOps

- Designed and delivered reusable backend components and shared services, accelerating feature delivery across engineering teams.
- Contributed to CI/CD pipeline configuration and deployment automation on Azure and AWS cloud platforms.
- Enhanced system observability through structured logging, monitoring dashboards, and automated alerting.

Persistent Systems Limited | Pune, India

Module Lead | 09/2015 – 02/2020

- Led architecture design and full-cycle software development for enterprise desktop and web applications in the Life Sciences domain (client: Agilent Technologies).
- Managed requirements gathering, technical design documentation, UI mockups, implementation, testing, and production release for multiple concurrent projects.
- Owned technical issue resolution, mentored junior developers, and maintained engineering knowledge-base documentation.
- Delivered Smart Alerts web application (C#, WCF, MongoDB, AWS) enabling configurable instrument monitoring, threshold-based alerting, and dashboard reporting.
- Built ACE (Agilent Compliance Engine) — WPF desktop application for pharmaceutical hardware compliance testing and automated report generation (C#, WPF, MVVM, WCF).
- Developed DDK (Domain Development Kit) — XML file management tool with intelligent validation rules, replacing error-prone Excel-based workflows (C#, WPF, Prism).

Tech Stack: C#, WPF, MVVM, WCF, ASP.NET Core, MongoDB, jQuery, JavaScript, AWS, SQL Server, T-SQL, WIX, Prism, IBM ClearCase

KEY PROJECTS

Agentic QA Framework

Technologies: LangGraph, Python, FastAPI, OpenAPI, Swagger

- Designed and built an AI-powered API testing framework enabling QA engineers to write test cases in plain English with zero coding or scripting.
- Framework ingests OpenAPI/Swagger JSON to auto-generate tool definitions; LangGraph agents dynamically invoke APIs, validate responses, and report results.
- Reduced QA onboarding time and eliminated dependency on manual test script maintenance.

Agentic AI Assistant

Technologies: Python, LangGraph, LLM Integration, REST API, FastAPI

- Developed a conversational AI assistant capable of multi-step task automation, API orchestration, and real-time enterprise service integration.
- Implemented modular agent architecture allowing plug-in extension of new capabilities without core system changes.

Intelligent Email Review System

Technologies: ML.NET, Python, Binary Classification, Multiclass Classification, Data Pipelines

- Built end-to-end ML classification system automating email triage: binary model for relevance filtering, multiclass model for business category assignment.

- Delivered full ML pipeline (ingestion, preprocessing, feature engineering, training, evaluation, deployment) with results visualised via interactive bar charts.
- Eliminated manual review bottleneck, significantly improving processing throughput.

Compliance Auditing System — Compliance Accelerator

Technologies: C#, .NET Core 5/6, Elasticsearch 7.10, Angular, SQL Server, T-SQL, SQLite, WIX

- Developed audit log capture system tracking all data changes in SQL Server and indexing into Elasticsearch for compliance reporting and search.
- Implemented Elasticsearch query optimisation and index management for high-performance retrieval at scale.

EDUCATION

Bachelor of Engineering (B.E.) — MIT Aurangabad (2015)

Diploma in Information Technology — CSMSS Aurangabad (2012)

CERTIFICATIONS

Microsoft Certified: Azure Developer Associate (AZ-204) — Microsoft